

Lec 27 Conditional VAE

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Transcript

[00:00] Hello everyone. Welcome to the lecture series on fundamentals of generative AI and large language models. We are in the week four and week four is all about advanced VAE concepts. So in this a particular session, let us have a quick recap of VAE and then we will look into the most important called a conditional VAE and we will look into its architecture and what exactly is the loss function of VAE. So we have already seen in our previous week the entire working mechanism of VAE and we have also seen the loss function. How exactly the loss function, what is the difference between VAE and conditional VAE loss function and as always we'll understand the architecture towards the end with one a numerical example. So

[00:46] let's get started with a quick recap of a VAE. So this is what is the architectural diagram that we have seen where if you look into this is a standard VAE architecture. How how can you say it as a standard VAE? Because the model is been mainly trained only with an input data, right? If you see so far for an model we were we have passed only the input to data and here that input to data is X. And what is our ultimate aim? The aim of VAE is that it has to learn this compressed latent representation. When an input is a passed to an encoder, encoder has to learn the compressed

representation of the input and that is latent representation Z . And that Z when you pass it to an decoder, the main aim is to reconstruct the input

[01:42] or generate the data and here that data we call it as an $\hat{X}$. So this is what is the VAE overall. If you're looking into in detailed architecture, so what is that we are doing? We are first passing the input and that input is given to an encoder. And what is the role of an encoder here? Encoder converts this input, whatever you are passing, into an probabilistic latent representation. What do you mean by probabilistic latent representation?

[02:14] Here, ultimately, it is mapping to input X to a distribution over the latent representation, which means the uh encoder is mainly aiming to produce the mean and then the log variance of what? Of appropriate posterior distribution. So, once the encoder gives this output, then which means once the encoder gives the mean and then the variance of an approximate posterior distribution, then we are going to sample Z . So, we are going to get this Z , right? How is that the Z is obtained?

[02:54] Z is a sampled using reparameterization trick. What is reparameterization trick? We take the mean, standard deviation, and then the epsilon. And epsilon is being sampled from a normal distribution. So, once we get the latent representation Z , Z is a passed as an input to an decoder. So, decoder takes a latent variable Z as an input and reconstructs or generates the data. Which means it is mainly aiming to map the latent vector Z back to the original data space, right? So, the function that we are representing, this entire is denoted with the function the probability of getting an generated data X given the latent representation Z . So, what exactly we have seen?

[03:47] Standard VAE mainly aims to learn from the input. It has to get the Z . From Z , it has to get the reconstructed $\hat{X}$. Right? This is what uh is uh the uh VAE. Now, let us look into what is the need for uh the uh next architecture, which is a point of discussion for this particular session, which is conditional VAE. Why exactly we require this? That means there has to be some limitation with respect to existing one. What is that limitation? If you look into a standard VAE, generation is not explicitly controlled. What do you mean by generation is not explicitly controlled? Now, you are passing an input to Z , and Z is producing an you know output. When Z is passed to a decoder, decoder is getting the output.

[04:39] So, now, what if you have a condition like this? So, that's what explicitly it is not explicitly controlled. Now, what if you want to generate uh um a specific type of the um output? Um I want a specific type of an output. Probably, you know, generate only the image uh five, or generate only the cat images, generate only the dog images. If you're working in medical application, what if you have a less number of malignant images? Generate Now, you want a controlled output. What if you want to generate a specific output? Then, we need to give uh extra guidance. We need an extra guidance

if you want to generate a specific type of the output.

[05:23] What exactly mean by this? Suppose, if you want to generate an image of five, or if you want to generate an a cat image, then, you need an extra guidance. So, for that, we require an conditional VAE. What is conditional VAE? It is exactly an extension of VAE. Working mechanism will remain the same as VAE, but we give an extra input. What is that extra input? You see, for the input we are passing the label also. And to the encoder you are passing both X and Y .

[05:56] And when you you get a Z . When you get a Z , when you pass a Z to the decoder, you are again passing this. So, which means this encoder and a decoder, you know, we are providing an additional input. So, that additional input is a condition here. And that condition is denoted it as a Y . And this Y can be a class label or it can be an attribute. And look at this reparameterization trick. How is that we are writing? Ultimately, we were before we were writing it as, you know, of X and σ of X and ϵ . We were taking only so much. But here, along with the input X , we are also passing a label Y .

[06:46] Therefore, it is X , Y and here it is X , Y . So, a little more if you have to understand, look at this particular architectural diagram. For the encoder, input X and Y is being passed. So, now whatever this μ and variance we are getting, is it not which will have X and Y as an impact? X and using X and Y only you are getting this mean and then the variance. Therefore, this is what you have X and Y here. So, I hope you have understood what is conditional VAE.

[07:26] Conditional VAE is an extension of VAE where we provide an additional input. So, now you can ask the question, why we need to provide an additional input? What is that by providing an additional input we are getting an benefit? Now, you just recall, go back and then see the architecture of VAE. In the normal VAE or in the standard VAE, decoder receives this entire is the decoder. Decoder is receiving only Z as the input, right? Then if decoder is receiving a Z Z is the latent representation, which must store the you know information of the image and it should also store the class information.

[08:13] Correct? Do you agree to this? Suppose if you are passing an handwritten digit three, then Z should capture the style of the image. How is three written? How is the curve? Whether it is thin or thick, slant, sharp. It has to capture those feature information and it should also capture the class information, which means it belongs to handwritten digit three. So, Z has to capture both of these information when it comes to standard VAE. That is if an image handwritten three, then Z must somehow encode that it is a three, right? So, but whereas in conditional GAN in a C in conditional VAE in a CVAE, if you look into » [snorts] » what is that the you are providing to the model, is it not the input Y is also

[09:09] passed to Z . Y is also passed to to Z . So, when you are passing it to the

decoder along with the features in the image, you are providing the class label also, which means the model need not have to bother about to which class it belongs to or the class information. So, can we think in this way? Somebody is already providing that information. If that is the case, I need to focus only on the important the features or the meaningful features of the image. So, is it not this is a more efficient when comparing to VAE?

[09:50] In VAE, we were passing only Z. Z has to capture both the information, information about the image as well as the class information. But in a CVAE, conditional VAE, we are not model We do not have to bother about the class information. You're explicitly passing it. If that is the case, can it focus only on capturing the information from the image? Is it not you're putting less burden on a Z? Z Now, Z is more efficient in capturing. Did you get this intuition? So, this is all is you know, the advantage of going for conditional.

[10:31] So, decoder here receives both X and Sorry, Z and Y. » [snorts] » So, Y already tells the decoder the class that it belongs to. So, because of this Z can focus more on variations within that class. So, Z can capture the styles such as whether it is a thin or a thick digit, whether it is a tilt or a straight image, whether it is a tall or short digit, whether it is curved or sharp writing style. All that is what Z can capture without actually bothering to which class it belongs to because explicitly you're passing the class information. So, therefore What What is that the model learns here?

[11:18] The model learns from X and Y is given to Z. And from this, you're getting a reconstructed one. So, for Z, again you will be passing label Y. So, with this, you see, encoder uses both X and Y. Earlier in standard VAE, we just had this. This is the standard VAE. But in conditional VAE, you you are passing X and Y both. And Z as well as Y. So, decoder uses earlier it was X given only Z. But here, along with the Z, you are also passing Y. So, what is that we are doing? For encoder and decoder uses these inputs so that it can produce the controlled output now. That was the problem with the standard VAE.

[12:18] We had a problem that you cannot generate a controlled output. Now, because you're passing the label, is it not at encoder you're passing a label? At for encoder also the label is passed. For a decoder also the label is passed. So, because of this, it can produce the controlled output. And these inputs, this Y what you are additionally passing, ensure the model generates the data according to the given condition only. Therefore, you're increasing the flexibility.

[12:47] Definitely the flexibility has been increased because now whatever the model is a generating the data, it is generating according to the given condition. So, therefore, there is an controlled generation in CVAE. So, I hope theoretically you have understood the concept of conditional VAE. It's as simple as this. To the original model, how along with the input you are passing, you are also passing Y. And to a

decoder, along with the Z , we are also passing Y .

[13:23] So, with this, there is a controlled output and it provides some more flexibility. So, if this is the case, what is the loss function in CVAE? So, the whatever you are seeing uh now, this is the standard VAE architecture. This is a standard VAE loss function. This is the loss function of standard VAE. Uh this we have seen in detail in our previous week only. Now, what we are discussing is the conditional VAE. So, how should be the loss function? Is there any major change? Change is only along with for an encoder, we are also passing Y , and for a decoder, we are also passing Y . So, that means for this, you are passing Y , and for a decoder also, you are passing Y . So, is it not the change is only so much? And

[14:14] of course, here also, you are passing Y . So, uh additional input is being passed to a conditional VAE, so that you will have a controlled output generation, and it enhances the flexibility of using this particular model. Now, so this is the overall loss function. Now, let us try to recall from the numerical example that we have solved. So, before we go into the next slide, that is the numerical example, I want all the listeners here to go back to our week three, and then solve the problem.

[14:54] If you if you have already solved that, try to recall that concept. If you have still not solved, you can watch that particular week three last numerical session. So, there, what was the input that we had passed? Is it not this is the input that we have considered in a normal VAE, right? Or in a standard VAE. Let me not you know, don't get confused when I say normal, it is a standard VAE. So, this is the input that we had passed.

[15:21] Suppose, this input belongs to a particular class. Class information also should be passed. Now, suppose if it belongs to some binary class. If it belongs to an a binary class, then the label will be either that class or this class. So, it will be either zero or one. Or if you're using one hot encoding, it will be 1 0 or it is 0 1. So, I have used one hot encoding and then if it belongs to one particular class and that class label is denoted as 1 0.

[15:58] That 1 0 indicates, suppose if you if you take this as an Iris data set and for example, if you consider Iris data set is having two classes. Although original is having three, but just for understanding, let us assume it is having two classes. Iris setosa, Iris versicolor. If that is the case, this sample belongs to only one class. If it belongs to one class, then it is Iris setosa or it is versicolor. So, if that is the case, if you're using one hot encoding, it belongs to one class, it does not belongs to another class. So, that is the meaning of 1 0 here.

[16:37] Now, since it is an conditional VAE, what is the input that you are passing? Along with the input, the class label also should be passed as an input. Now, you see, along with the input, this entire is X . Along with that, you are also passing Y . Right? Now, earlier in our standard VAE, only input was passed. Now, along with the input,

you're also passing the class label. Now, can you recall what will be our next concept? Is it not you'll be multiplying it with the weight and then rest everything will remain the same.

[17:14] You'll be using an additional weights here. Why additional weights? Because there earlier it was a four features, so four input. Now it is a six. So definitely weights when you are establishing a connection, you require two more additional weights. So now after you perform the entire calculation, then let us assume you got the hidden representation. How is that you get the What was the hidden representation that you have obtained in standard VAE? Is it not this was the hidden representation? This is the same example what we had solved. So from that example, we know that VAE first hidden representation has given this particular output. 1.07, 1.93, and 0.87. This is the output that we have obtained. Now after

[18:04] we pass this particular input and for some weights that we have taken, let us assume conditional VAE after adding this label, this is the you know, output that we have obtained from the hidden representation. Now can you look into intentionally have shown both so that you will have a clear understanding. Can you see the difference between hidden representation of VAE and the CVAE, conditional VAE? Definitely in conditional VAE, hidden representation is carrying additional information, right? There is comparing to this, we have the weights the values are little high here. What is that? So because the labels the class label has been added, this label contributes additional weighted information. Definitely

[18:54] to this encoded hidden layer, there is some information more information is there comparing to this particular hidden representation. So what is that I'm trying to explain you here? Along with the input, we are also passing the class label. Y. Because you're passing the class label Y, there is an additional information you're providing. How is that you can able to understand what is the role of this additional information? Look at this. Definitely, comparing to the normal standard VAE, the CVAE has more information. Why this more information has come? It is because of this class label.

[19:36] I hope this is clear to you. Now, once this is done, what what should be our next step? You have to pass that Let us assume from there you have obtained the latent I mean distribution parameters, mean and variance. From there, you have to sample Z. Okay, let us assume the Z is already been sampled. If it was in a normal or a standard VAE, Z was only so much. Right? But now, you're not just passing a Z, you have to pass a Z and a Y, both.

[20:12] So, because this is Z which has been obtained by passing the previous output. By passing this value, we have obtained this Z. We have obtained this. Now, to the decoder, you're just not passing a Z, you have to pass even the label also. What is the label that we have used? The class label 1 0. So, along with the Z, this is just a Z,

along with the Z , you're also passing a Y . Right? That is what along with the Z , you're also passing Y .

[20:45] I hope you are getting this. So, now what is this Z ? Is it not an input to the decoder? So, go back to the architectural diagram. To the Z , you're also passing a Y . So, is it not we have done this? Right? So, now what exactly this information tells to the decoder? Generate or reconstruct X using this latent information Z . Uh is it not uh when you pass this is Z to an decoder, ultimately decoder has to give this uh generated or constructed X or $\hat{X}$. And it has to generate using this latent information Z , but it is just a not a Z , it is a now even conditioned on class label Y .

[21:33] So, this is what is the major difference. So, you are giving a Z , but along with Z , Y is also being given. So, this is how the information is given to a decoder. Now, at a decoder side, you go back and then recall from what we have solved, we have obtained the reconstructed data $\hat{X}$. So, here also you see this was the data that we had obtained at a um you know, for a standard VAE. But here because it is an conditional VAE, because some additional input is being passed, definitely there is a change in the reconstructed or generated data.

[22:14] So, over a period of time what will you do? You will calculate the loss, you will back propagate, and then you will update the weights and then the bias. Over a period of time, definitely there will be an improvement in whatever the reconstructed data that we have obtained. This, when you compare it with the original, it will be very much matching. So, this uh shows the label Y changes the decoder hidden representation, definitely. So, because of this, you can see there is a change in the reconstructed output also. Because there is a because of Y , there is a change in the hidden representation, and because of change in the hidden representation, there is change in Z . Because of change in Z , there is a change in a

[22:56] reconstructed data also. So, this is all about uh VAE. In the next particular session, we will look into a latent space interpolation. This is one of the most important concept when we are generating the data. So, what is that we have looked into this particular session? This session mainly focused on what exactly is VAE. We have seen a quick recap, and we have seen an extension version of VAE called conditional VAE, which is more flexible when compared to a normal or a standard VAE. With this, we have come to an end of this particular session. Thank you.